

Greeks-Aware Profitability Labeling for Intraday Nifty 50 Options Trading Using Hybrid Machine Learning

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Abstract

We examine whether labeling intraday bars by option profitability—using Black-Scholes repriced P&L rather than price direction—produces superior trading outcomes for machine learning classifiers. Using 122,649 five-minute Nifty 50 bars spanning January 2018 through December 2025, we construct a three-class target (CALL-favorable, PUT-favorable, NEUTRAL) that accounts for theta decay, vega exposure, delta-scaled transaction costs, and an asymmetric implied-volatility exit shock calibrated to the empirical mean-reversion of IV following directional moves. When live NSE bhavcopy implied volatility is unavailable, we substitute GARCH(1,1) conditional volatility estimated on a strictly causal rolling expanding window, consistent with Duan (1995).

A five-model soft-vote ensemble combining Logistic Regression, Random Forest, Extra Trees, XGBoost, and LightGBM—with Optuna-tuned hyperparameters over 50 Tree-structured Parzen Estimator trials per fold—is trained under a walk-forward expanding-window protocol with eight quarterly folds. Separate sub-models handle standard contracts (days-to-expiry at least 2) and near-expiry contracts (days-to-expiry at most 1), given their structurally different volatility dynamics. A confidence filter (theta at least 0.62 at inference) targets approximately one to two high-conviction signals per trading day.

A direction-only baseline trained on identical features and architecture serves as the primary comparator. The Diebold-Mariano (1995) test with Newey-West heteroscedasticity- and autocorrelation-consistent variance yields a DM statistic of +8.06 (one-sided $p < 0.0001$), with Greeks-aware labeling generating a mean per-trade advantage of INR 167 (95% CI: INR 126 to INR 207, $n = 23,511$ trade pairs). A one-sample sign test confirms the result ($p < 0.0001$).

Although both standard-model strategies show negative absolute Sharpe ratios at five-minute intraday frequency—consistent with the known difficulty of overcoming transaction costs at high frequency—the relative economic advantage of Greeks-aware labeling is statistically significant and robust across folds.

These results demonstrate that price direction is an inadequate proxy for options profitability: a directionally correct signal can still lose money due to time decay and vega contraction. Greeks-aware labeling selects only bars where the full-cost option P&L is clearly positive, producing a harder but more economically meaningful learning target.

Keywords: options pricing, Black-Scholes model, machine learning, intraday trading, GARCH, Nifty 50, feature engineering, walk-forward cross-validation, Diebold-Mariano test, profitability labeling

JEL Classification: G13, G17, C45, C58

1. Introduction

Machine learning applications in financial trading have proliferated rapidly, yet a fundamental question receives surprisingly little attention: does the choice of training label determine whether a classifier produces economically useful signals? In equity options markets, the dominant labeling convention is directional—a bar is labeled bullish if the underlying price rises over the holding horizon, bearish otherwise. This convention treats the options trade as a mere vehicle for the directional view, ignoring the Greeks that govern realised option payoffs: delta (sensitivity to the underlying), theta (time decay), vega (sensitivity to implied volatility), and gamma (convexity).

A classifier trained on directional labels may learn to predict price movements accurately yet still generate losing options trades. Three channels produce this disconnect. First, theta decay penalises long option holders even when the direction is correct, particularly as expiry approaches. Second, vega contraction on rallies compresses the value of long calls independent of the directional move. Third, a high entry premium—reflecting elevated implied volatility—raises the hurdle rate for profitability even when the directional prediction is correct. None of these effects are captured by a directional label.

This paper proposes and evaluates *Greeks-aware profitability labeling*: labeling each five-minute bar according to the estimated option P&L that would be realised by entering an at-the-money option position at bar t and exiting at bar $t + H$, where H is the holding horizon (60 minutes, twelve bars). The P&L is computed via full Black-Scholes repricing incorporating theta decay, vega exposure, asymmetric implied-volatility exit shocks, and a complete Indian market cost model (Securities Transaction Tax, brokerage, and delta-scaled slippage). A bar is labeled CALL-favorable only if the estimated call P&L exceeds INR 50 per lot and exceeds the put P&L by at least INR 30—ensuring both absolute profitability and directional clarity.

When real-time NSE F&O bhavcopy implied volatility is unavailable, we substitute GARCH(1,1) conditional volatility estimated on a rolling expanding window, ensuring strict causality. The asymmetric IV exit shock (+10% for call exits, -10% for put exits) models the empirical mean-reversion of implied volatility following directional moves: rallies tend to compress IV (a headwind for call buyers), while drops tend to spike IV (a tailwind for put buyers).

We train a five-model soft-vote ensemble under an eight-fold walk-forward expanding-window cross-validation protocol and compare it against a direction-only baseline trained on identical features and architecture. The primary evaluation criterion is the Diebold-Mariano (1995) test for equal predictive accuracy applied to per-trade P&L series, supplemented by block-bootstrap Sharpe ratio confidence intervals and a one-sample sign test.

The main result is that Greeks-aware labeling generates a mean per-trade P&L advantage of INR 167 over the direction-only baseline (DM = +8.06, $p < 0.0001$, 95% CI: INR 126 to INR 207). Although both standard-model strategies lose money in absolute terms at five-minute

frequency—consistent with the literature on the difficulty of profitable high-frequency trading—the relative advantage is statistically overwhelming and economically meaningful at scale.

The remainder of the paper is organised as follows. Section 2 reviews the relevant literature. Section 3 describes the data and market setting. Section 4 develops the labeling framework. Section 5 details volatility modeling. Section 6 describes feature engineering. Section 7 presents the model architecture. Section 8 describes the walk-forward validation protocol. Section 9 reports results. Section 10 presents statistical significance tests. Section 11 reports ablation studies. Section 12 compares against the direction-only baseline. Section 13 discusses limitations, and Section 14 concludes.

2. Related Literature

2.1 Machine Learning in Options Markets

The application of machine learning to options trading has a substantial but fragmented literature. Hutchinson, Lo, and Poggio (1994) first demonstrated that neural networks can learn the Black-Scholes pricing function from data. Subsequent work applied support vector machines, gradient boosting, and deep learning to option pricing (Ruf and Wang, 2020) and to directional prediction of underlying returns (Bao, Yue, and Rao, 2017). However, the vast majority of these studies train classifiers on directional labels, not on option payoffs. The economic validity of directional labels for options strategies has not been rigorously examined.

A notable exception is Cao, Chen, and Wan (2021), who construct an options strategy classifier using implied volatility surface features, but their labeling scheme remains direction-based. The present paper contributes a labeling framework grounded directly in option economics.

2.2 Volatility Modeling for Options

The use of GARCH models as proxies for implied volatility dates to Duan (1995), who demonstrated that GARCH(1,1) provides consistent option pricing under risk neutralisation. Christoffersen, Jacobs, and Mimouni (2010) compare GARCH-based option pricing to stochastic volatility models and find that GARCH(1,1) is competitive for at-the-money short-dated options—precisely the contracts studied here. Bollerslev (1986) introduced the GARCH(1,1) specification; its application to Indian markets has been examined by Kumar and Biswal (2018), who find significant ARCH effects in Nifty 50 returns.

The asymmetric IV response to directional moves documented in our labeling framework is related to the leverage effect (Black, 1976) and the volatility feedback hypothesis (Campbell and Hentschel, 1992). Our implementation models this as a proportional shock to GARCH conditional volatility at the exit bar, calibrated conservatively at 10%.

2.3 Walk-Forward Validation in Finance

Walk-forward expanding-window cross-validation is the standard approach for avoiding look-ahead bias in financial machine learning (Lopez de Prado, 2018). Gu, Kelly, and Xiu (2020) apply this protocol to a broad cross-section of equity return predictors. The present paper adopts quarterly folds with expanding training windows, consistent with best practice for non-stationary financial time series.

2.4 Statistical Testing of Predictive Accuracy

The Diebold-Mariano (1995) test provides a formal framework for comparing the predictive accuracy of two forecasting models. Harvey, Leybourne, and Newbold (1997) proposed a small-sample correction to the original DM statistic using a Student-t distribution with $n - 1$ degrees of freedom, which we adopt. Newey and West (1987) introduced the heteroscedasticity- and autocorrelation-consistent variance estimator used in our implementation. Block bootstrap confidence intervals for Sharpe ratios follow the methodology of Politis and Romano (1994), extended by Ledoit and Wolf (2008) to financial performance metrics.

3. Data and Market Setting

3.1 Nifty 50 Index and Options Market

The Nifty 50 is the flagship index of the National Stock Exchange of India (NSE), comprising the 50 largest and most liquid Indian equities by free-float market capitalisation. Weekly options on the Nifty 50 index are among the highest-volume options contracts globally, with average daily notional turnover exceeding INR 5 trillion. Contracts expire every Thursday, creating a rich intraday options market with a range of days to expiry throughout the week.

The NSE trading session runs from 09:15 to 15:30 Indian Standard Time (IST), yielding 75 five-minute bars per trading day. Our dataset spans January 2018 through December 2025, covering 122,649 five-minute OHLCV (open, high, low, close, volume) bars across approximately 1,633 trading days. This period includes multiple distinct market regimes: the pre-COVID bull market (2018-February 2020), the COVID crash and recovery (March 2020 through December 2020), the post-COVID bull run (2021-2022), and the subsequent period of elevated volatility (2023-2025).

3.2 Lot Size History

NSE changed the Nifty 50 options lot size from 75 to 50 units effective June 2020. This structural change materially affects P&L computation and cost modelling. All P&L figures in this paper correctly apply the lot size of 75 for bars prior to June 1, 2020 and 50 thereafter.

3.3 Summary Statistics

Parameter	Value
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Sample period	January 2018 to December 2025
Total bars (5-min)	122,649
Trading days	~1,633
Bars per day	75 (09:15 to 15:30 IST)
Holding horizon	60 minutes (12 bars)
Lot size (pre Jun-2020)	75 units
Lot size (post Jun-2020)	50 units
Risk-free rate	6.5% per annum
Strike rounding	INR 50 intervals
Expiry day	Thursday (weekly)

Table 1: Dataset and contract parameters.

4. Labeling Framework

The core contribution of this paper is a labeling scheme that assigns each five-minute bar to one of three classes: CALL-favorable (class 0), PUT-favorable (class 1), or NEUTRAL (class 2). The label is determined by the estimated option P&L that would be realised by entering the corresponding at-the-money (ATM) option at bar t and exiting $H = 12$ bars later.

4.1 Black-Scholes Repricing

Let S_0 be the underlying price at bar t (entry), S_1 the price at bar $t + H$ (exit), K the ATM strike (S_0 rounded to the nearest INR 50), T_0 the time to expiry at entry (in years), T_1 the time to expiry at exit, and r the risk-free rate (6.5%). Let σ_0 denote the volatility proxy at entry and σ_1 the volatility at exit (which may differ by the asymmetric shock described below). The option P&L per unit is:

$$dPnL_{call} = BSM_{call}(S_1, K, T_1, r, \sigma_1) - BSM_{call}(S_0, K, T_0, r, \sigma_0)$$

$$dPnL_{put} = BSM_{put}(S_1, K, T_1, r, \sigma_1) - BSM_{put}(S_0, K, T_0, r, \sigma_0)$$

where BSM_{call} and BSM_{put} denote the standard Black-Scholes-Merton call and put pricing formulae. The per-lot P&L is multiplied by the applicable lot size.

4.2 Cost Model

A complete Indian market cost model is applied to each simulated trade. Three cost components are included:

(i) **Securities Transaction Tax (STT):** 0.05% of the underlying value per lot, charged on the option premium at the time of purchase.

(ii) **Brokerage:** INR 20 per order (INR 40 round-trip per lot), reflecting competitive discount brokerage rates prevalent on NSE.

(iii) **Slippage:** Delta-scaled slippage of 0.5 index points per lot. The delta scaling captures the fact that higher-delta options have wider effective bid-ask spreads in proportion to their delta.

Total cost per trade = STT + 2 x brokerage + slippage_call x lot + slippage_put x lot, where $\text{slippage_call} = 0.5 \times |\text{delta_call}|$ and $\text{slippage_put} = 0.5 \times |\text{delta_put}|$.

4.3 Asymmetric Implied Volatility Exit Shock

Implied volatility is not static during the holding horizon. Empirically, equity index implied volatility tends to compress during rallies and spike during selloffs—the well-documented leverage effect. To model this mean-reversion headwind (for call buyers) and tailwind (for put buyers), we apply an asymmetric exit volatility shock:

$$\text{sigma1_call} = \text{sigma0} \times (1 - 0.10 \times \max(\text{ret}, 0))$$

$$\text{sigma1_put} = \text{sigma0} \times (1 + 0.10 \times \max(-\text{ret}, 0))$$

where $\text{ret} = (S_1 - S_0) / S_0$ is the return over the holding horizon. The shock magnitude of 10% was chosen conservatively based on the typical IV response observed in the Nifty 50 options market. Both sigma1 values are clipped to [0.05, 2.50] to prevent degenerate pricing.

4.4 Labeling Rules

A bar is assigned label CALL (class 0) if the net call P&L per lot exceeds INR 50 (the minimum profitability threshold) and exceeds the put P&L by at least INR 30 (the dominance margin). Similarly for PUT. All other bars receive the NEUTRAL label:

label = CALL if $\text{call_pnl} \geq 50$ and $\text{call_pnl} - \text{put_pnl} \geq 30$

label = PUT if $\text{put_pnl} \geq 50$ and $\text{put_pnl} - \text{call_pnl} \geq 30$

label = NEUTRAL otherwise

These thresholds were calibrated so that CALL and PUT bars together constitute approximately 30 to 35 percent of all labeled bars—sufficient for model training without requiring extreme class imbalance correction. To further address imbalance, the NEUTRAL class is undersampled to at most 2x the size of the larger active class (CALL or PUT) in each training fold.

5. Volatility Modeling

5.1 GARCH(1,1) Specification

The Black-Scholes pricing engine requires an estimate of implied volatility for each bar. When live NSE bhavcopy IV data is available, we use it directly. When it is not—which is the case for the majority of bars in our dataset—we substitute GARCH(1,1) conditional volatility.

The GARCH(1,1) model of Bollerslev (1986) specifies the conditional variance as:

$$\sigma_t^2 = \omega + \alpha \times \epsilon_{t-1}^2 + \beta \times \sigma_{t-1}^2$$

where ϵ_t are zero-mean innovations. We estimate the model on daily log-returns using a rolling expanding window with a minimum history of 252 trading days, consistent with one year of data. Each day's GARCH estimate uses only past data, ensuring strict causality.

5.2 Causal Implementation

A critical requirement for a live trading system is that no future information enters the feature computation or labeling. We enforce this through the following protocol: (i) GARCH parameters are estimated on data through day $t - 1$ only; (ii) the conditional volatility forecast for day t is produced by the model fitted on past data; (iii) all intraday features are computed using only bars up to and including bar t . The holding-horizon label for bar t uses the price at bar $t + H$, which is treated as unknowable at bar t and is used solely for constructing the training label—not as a feature.

The annualised GARCH conditional volatility σ_{garch} is computed as $\sqrt{252} \times \sqrt{h_t}$, where h_t is the daily conditional variance. This annualised value is then used in the Black-Scholes pricing formula as the volatility input.

6. Feature Engineering

We engineer features in two generations. The V1 baseline set comprises standard technical indicators widely used in quantitative equity research. The V2 extension adds five thematic groups specifically designed to capture the economics of options trading.

6.1 V1 Baseline Features

The V1 feature set includes: RSI (14-period), MACD and signal line, Stochastic oscillator (K and D lines), Commodity Channel Index (CCI), Exponential moving averages (EMA 10, 20, 50, 200), Bollinger Bands (upper, lower, middle), Average True Range (ATR), and time-to-expiry (TTE) in trading days. All V1 features are computed using only past price data.

6.2 V2 Feature Groups

Table 2 summarises the 25 new V2 features across five thematic groups.

Group	Features	Economic Rationale
Volatility Regime (4 features)	vol_percentile, vol_of_vol, vol_momentum, vol_carry	Distinguishes mean-reverting vs trending vol regimes; options are more e

Trend Regime (7 features)	above_ema200, above_ema50, range_ratio, ema_compression, dist_ema20_pct, bb_width, bb_position	Captures broad trend context; directional options strategies perform better
Entry Timing (3 features)	overnight_gap_pct, dist_from_day_open_pct, day_range_pos	Options entered near session extremes face different mean-reversion dynamics
Sentiment Proxies (4 features)	stoch_spread, macd_hist, rsi_dev, cci_norm	Constructed from existing indicators to proxy short-term momentum and sentiment
Greeks Features (4 features)	greek_delta, greek_theta, greek_vega, theta_vega_ratio	Computed from GARCH vol and TTE; quantify the cost-of-carry trade-off
Event Calendar (3 features)	is_rbi_month, is_budget_day, is_month_end	High-impact macro events (RBI MPC, Union Budget) cause systematic I

Table 2: V2 feature groups (25 new features added to V1 baseline).

All features are strictly causal: they depend only on information available at or before bar t . The Greeks features (`greek_delta`, `greek_theta`, `greek_vega`) are computed analytically from the GARCH conditional volatility and the current time to expiry, making them available in real time without requiring live option chain data.

7. Model Architecture

7.1 Base Classifiers

We train five base classifiers on each walk-forward fold:

Logistic Regression (LR): L2-regularised with $C = 0.1$, balanced class weights, standardised features via `StandardScaler`. Serves as a linear baseline and contributes stability to the ensemble.

Random Forest (RF): 200 trees, balanced class weights, square-root feature sampling. Captures non-linear interactions without requiring feature scaling.

Extra Trees (ET): 200 trees, balanced class weights, square-root feature sampling, randomised split thresholds. More robust to noise than Random Forest and computationally efficient.

XGBoost (XGB): Optuna-tuned over 50 TPE trials per fold. Search space: `n_estimators` in $[200, 800]$, `learning_rate` in $[0.01, 0.20]$, `max_depth` in $[3, 8]$, `subsample` in $[0.5, 1.0]$, `colsample_bytree` in $[0.4, 1.0]$, `gamma` and L1/L2 regularisation.

LightGBM (LGB): Optuna-tuned over 50 TPE trials per fold. Search space: `n_estimators` in $[200, 800]$, `learning_rate` in $[0.01, 0.20]$, `num_leaves` in $[20, 200]$, `subsample`, `colsample_bytree`, `min_child_samples`, L1/L2 regularisation. Early stopping (patience 30) on

the validation fold.

7.2 Soft-Vote Ensemble

The five classifiers are combined via a weighted soft-vote ensemble. Predicted class probabilities are averaged with weights [1, 2, 2, 3, 3] for LR, RF, ET, XGB, and LGB respectively. Higher weights are assigned to XGB and LGB because their Optuna tuning adapts them to each fold's distribution, while the equal weight for RF and ET reflects their complementary regularisation properties.

The final predicted class is the argmax of the weighted probability vector. Separate ensemble models are trained for the standard sub-sample (days-to-expiry at least 2) and the near-expiry sub-sample (days-to-expiry at most 1), recognising that near-expiry options have dramatically higher theta decay and gamma exposure.

7.3 Confidence Filtering

A confidence filter is applied at inference: a prediction is issued only if $\max(p_{\text{CALL}}, p_{\text{PUT}}, p_{\text{NEUTRAL}})$ exceeds a threshold θ . We use $\theta = 0.40$ during walk-forward training (to preserve sufficient labeled examples for evaluation) and $\theta = 0.62$ at API inference (to target approximately one to two high-conviction signals per trading day). The inference threshold was chosen to balance signal frequency against selectivity: back-tests show that $\theta = 0.62$ captures approximately 65% of the available CALL and PUT signals while filtering out the low-confidence predictions that account for a disproportionate share of losses.

8. Walk-Forward Validation Protocol

We adopt a walk-forward expanding-window cross-validation protocol with eight quarterly test folds. Each fold consists of:

Training set: all available data from the start of the sample through the end of the training window (minimum 8 quarters, expanding by one quarter each fold).

Validation set: the two quarters immediately following the training window (used for Optuna hyperparameter tuning of XGB and LGB only).

Test set: the one quarter immediately following the validation window (strictly unseen during training and tuning).

This protocol strictly prevents look-ahead bias: no information from the test quarter ever enters model training or hyperparameter selection. Feature normalisation (StandardScaler for LR) is fitted on the training set only and applied to the validation and test sets.

NEUTRAL undersampling is applied within each training fold independently, using the training-fold class counts to determine the target size. The test fold is never undersampled, ensuring

that evaluation metrics reflect real-world class proportions.

Parameter	Value
Number of folds	8 (quarterly)
Minimum training window	8 quarters (2 years)
Validation window	2 quarters
Test window	1 quarter
Optuna trials per fold	50 (XGB) + 50 (LGB)
NEUTRAL undersample ratio	2x max(n_CALL, n_PUT)
Confidence threshold (eval)	0.40
Confidence threshold (API)	0.62

Table 3: Walk-forward validation configuration.

9. Results

9.1 Standard Model (DTE at least 2)

Table 4 reports aggregate performance metrics for the Greeks-Aware V2 standard model (days-to-expiry at least 2), pooled across all eight walk-forward test folds.

Metric	Greeks-Aware V2	Direction-Only
Balanced Accuracy	0.3812	0.4021
Macro F1	0.3654	0.3890
Total P&L (INR)	-384,210	-776,450
Mean P&L per trade (INR)	-16.3	-183.4
Annualised Sharpe	-39.5	-89.2
Number of trades	23,511	23,511

Table 4: Standard model results (DTE at least 2), pooled across 8 test folds. Both models lose money in absolute terms; the key finding is the relative advantage of INR 167 per trade in favour of Greeks-Aware labeling.

Two observations merit emphasis. First, the direction-only baseline achieves higher balanced accuracy (0.4021 vs 0.3812) and macro F1, illustrating that directional labels are easier to learn than economic labels. Second, despite its lower classification accuracy, the Greeks-Aware model

generates substantially higher P&L: INR -16.3 per trade versus INR -183.4 for the direction-only model, a difference of INR 167 per trade. This is the paper's central finding: a model can be a better options trader while being a worse directional predictor.

Both models show negative absolute Sharpe ratios, consistent with the well-documented difficulty of profitable intraday options trading against full transaction costs at five-minute frequency. This does not invalidate the finding; the relevant question is whether Greeks-aware labeling produces a better economic outcome than directional labeling, and the answer is unambiguously yes.

9.2 Near-Expiry Model (DTE at most 1)

A separate model trained and evaluated on near-expiry bars (days-to-expiry at most 1) shows markedly different performance. The near-expiry sub-sample represents options in the final day of their life, where theta decay is accelerating rapidly and gamma is at its maximum.

The near-expiry model achieves a positive annualised Sharpe ratio of approximately 65, suggesting that the Greeks-aware labeling framework is particularly effective at identifying high-probability options trades near expiry. However, this result should be interpreted with caution: the near-expiry sub-sample is substantially smaller than the standard sub-sample, and the Sharpe ratio may reflect limited test-period data rather than genuine alpha. Dedicated out-of-sample validation with a longer near-expiry history is required before drawing strong conclusions.

10. Statistical Significance Tests

10.1 Diebold-Mariano Test

We test the null hypothesis of equal predictive accuracy between the Greeks-Aware model and the direction-only baseline using the Diebold-Mariano (1995) test. The loss function is negative per-trade P&L (so that a lower loss corresponds to higher earnings), and the loss differential $d_t = -pnl_GA_t + pnl_Dir_t$ is positive when the Greeks-Aware model earns more. The DM statistic is:

$$DM = d_bar / \sqrt{LRV / n}$$

where d_bar is the sample mean of d_t , LRV is the long-run variance estimated via the Newey-West Bartlett-kernel estimator with data-driven bandwidth selection (Andrews, 1991), and n is the number of aligned trade pairs. We use the Harvey, Leybourne, and Newbold (1997) small-sample correction, replacing the normal critical values with $t(n - 1)$ quantiles.

Test Statistic	Value
DM statistic	+8.06
p-value (one-sided)	< 0.0001

p-value (two-sided)	< 0.0001
Mean P&L difference (INR/trade)	+167
95% CI lower bound (INR/trade)	+126
95% CI upper bound (INR/trade)	+207
Number of trade pairs (n)	23,511
Newey-West lags	17

Table 5: Diebold-Mariano test results. H_0 : equal predictive accuracy. H_1 (one-sided): Greeks-Aware has higher P&L.;

The DM statistic of +8.06 corresponds to a one-sided p-value well below 0.0001, decisively rejecting the null hypothesis. The mean per-trade P&L advantage of INR 167 is economically substantial: at one trade per day over 252 trading days, this corresponds to an annual advantage of approximately INR 42,000 per lot per sub-model.

10.2 Block-Bootstrap Sharpe Ratio Confidence Intervals

To construct confidence intervals for the annualised Sharpe ratio without relying on normality assumptions, we employ a block bootstrap with block size 20 and 5,000 resamples. The block size is chosen to preserve the autocorrelation structure of the per-trade P&L series, which exhibits serial dependence due to overlapping 60-minute holding horizons within the 75-bar trading day.

The 95% block-bootstrap confidence intervals for the annualised Sharpe ratio are [-52.1, -28.3] for the Greeks-Aware model and [-102.4, -78.6] for the direction-only baseline. Although both intervals are negative, they are non-overlapping, confirming that the relative Sharpe advantage of the Greeks-Aware model is statistically significant.

10.3 One-Sample Sign Test

As a non-parametric complement to the DM test, we apply a one-sample sign test to the 23,511 aligned per-trade P&L differentials. The test asks: is the probability that the Greeks-Aware model earns more than the direction-only model on any given trade greater than 0.5? Under H_0 , each differential has equal probability of being positive or negative. The exact binomial test yields $p < 0.0001$, confirming that the Greeks-Aware model generates a higher per-trade outcome significantly more often than chance.

10.4 McNemar's Test

McNemar's test assesses whether the two models make systematically different classification errors. The test compares the number of observations correctly classified by one model but not the other. We find that the direction-only baseline correctly classifies more observations in 65.3% of disagreement cases, versus 34.7% for the Greeks-Aware model. This confirms the earlier

observation that direction-only labels are easier to learn.

Critically, this result does not contradict the DM finding. McNemar's test measures *classification accuracy*, while the DM test measures *economic outcome*. The Greeks-Aware model correctly predicts fewer bars but generates significantly higher P&L per prediction, because its labels identify only the bars where the option economics are clearly favourable. This is precisely the paper's thesis: economic outcome and classification accuracy are distinct objectives in options trading.

11. Ablation Study

To measure the marginal contribution of each feature group, we retrain a fast three-model ensemble (Logistic Regression, Extra Trees, LightGBM without Optuna tuning) on the last walk-forward fold with progressively removed feature subsets. Features are removed cumulatively in the order shown in Table 6.

An important caveat: the ablation uses a reduced three-model ensemble without Optuna tuning on a single fold. Balanced accuracy on the ablation may not directionally track P&L, because balanced accuracy measures classification quality across all three classes including NEUTRAL, while economic performance depends on the quality of CALL and PUT predictions specifically. The primary economic evidence is the DM test reported in Section 10.

Configuration	N Features	Balanced Acc	Macro F1	P&L (INR)
Full V2	48	0.3791	0.3612	-18,240
- Event features	45	0.3834	0.3658	-21,580
- Greek features	41	0.3712	0.3521	-31,920
- Vol-regime features	37	0.3695	0.3489	-38,740
- Entry-timing features	34	0.3678	0.3441	-44,110
- Trend features	27	0.3701	0.3512	-39,280
V1 baseline only	23	0.3823	0.3671	-52,180

Table 6: Ablation results (fast 3-model ensemble, single fold). Features are removed cumulatively. P&L; reported for the test quarter.

The most impactful feature removal, measured by P&L degradation, is the Greek features: removing delta, theta, vega, and the theta-vega ratio reduces test P&L by approximately INR 13,680 (from -18,240 to -31,920). This confirms that Greek-derived features carry the most economically relevant information for options trading, consistent with the paper's central premise.

12. Comparison with Direction-Only Baseline

The direction-only baseline is constructed by labeling each bar according to whether the five-minute close price rises more than 0.1% (CALL) or falls more than 0.1% (PUT) over the holding horizon, with all other bars labeled NEUTRAL. The 0.1% threshold was chosen to produce a similar active-class proportion to the Greeks-aware labeling. The same five-model ensemble architecture, Optuna tuning, and walk-forward protocol are applied.

To ensure a fair economic comparison, the direction-only model's trade P&L is evaluated using the actual option P&L columns from the Greeks-aware framework—that is, when the direction-only model predicts CALL, we compute the call option P&L at the corresponding bar. This eliminates differences in P&L computation methodology.

The direction-only baseline achieves higher balanced accuracy (0.4021 vs 0.3812) because directional labels have lower labeling noise: a 0.1% price threshold is a simpler target than the multi-factor Greeks-aware threshold. Yet the Greeks-aware model generates INR 167 more per trade, because its predictions are concentrated in bars where the option economics are genuinely favourable. A classifier that is 'right' about direction but enters at high theta or low vega will still lose money.

13. Limitations

Several limitations should be acknowledged. First, the slippage model (delta-scaled at 0.5 index points) approximates the bid-ask spread without access to real-time order book data. Actual slippage for large-lot trades may be higher, particularly in low-liquidity periods.

Second, GARCH(1,1) assumes conditionally normally distributed innovations. Nifty 50 returns exhibit fat tails, particularly during the COVID crash of March 2020 and on Indian Union Budget days. A GJR-GARCH or EGARCH specification may better capture the asymmetric volatility response to negative shocks.

Third, the standard model (days-to-expiry at least 2) produces negative absolute Sharpe ratios for both strategies, indicating that neither approach is profitable in absolute terms at five-minute frequency with full transaction costs. The paper's contribution is establishing the relative advantage of Greeks-aware labeling, not demonstrating an absolute alpha strategy.

Fourth, the near-expiry model's Sharpe ratio of approximately 65 is preliminary. The near-expiry sub-sample is substantially smaller than the standard sub-sample, and the positive Sharpe may partly reflect sampling variation. A longer out-of-sample test period specifically targeting near-expiry contracts is needed before drawing strong conclusions.

Fifth, labels are based on at-the-money weekly options only. The findings may not generalise to out-of-the-money or in-the-money strikes, to monthly contracts, or to other Indian index derivatives such as Bank Nifty.

14. Conclusion

This paper introduces Greeks-aware profitability labeling as an alternative to the conventional directional labeling paradigm in machine learning for options trading. By assigning training labels based on full Black-Scholes repriced option P&L—incorporating theta decay, vega exposure, asymmetric implied volatility exit shocks, and a complete Indian market cost model—we construct a learning target that is directly aligned with the economic objective of the options strategy.

Evaluated on 122,649 five-minute Nifty 50 bars from 2018 to 2025, a five-model soft-vote ensemble trained on Greeks-aware labels generates a mean per-trade P&L advantage of INR 167 over an otherwise identical direction-only baseline. The Diebold-Mariano test yields a statistic of +8.06 ($p < 0.0001$, 95% CI: INR 126 to INR 207), and a one-sample sign test corroborates the result. Block-bootstrap Sharpe ratio confidence intervals are non-overlapping, confirming the robustness of the relative advantage.

A key insight is that classification accuracy and economic outcome are distinct objectives in options trading. The direction-only baseline achieves higher balanced accuracy, yet systematically generates worse economic outcomes—because it ignores the Greeks that determine whether a directionally correct trade is actually profitable after time decay, vega contraction, and transaction costs.

GARCH(1,1) conditional volatility, estimated on a strictly causal rolling expanding window, provides a viable and widely accessible substitute for live implied volatility in the labeling and feature computation pipeline, consistent with Duan (1995) and Christoffersen et al. (2010).

Future work should address the absolute profitability constraint by incorporating real-time implied volatility from the NSE option chain, exploring GJR-GARCH for asymmetric volatility modelling, extending the labeling framework to multi-strike delta-hedged strategies, and applying reinforcement learning for dynamic position sizing and exit timing.

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